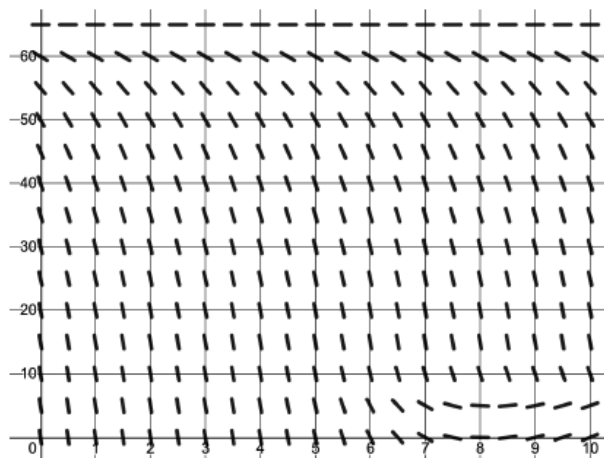


- 14 In the movie “The Matrix Reloaded”, Trinity is fighting an Agent and jumps out of the sixty fifth floor. (10 marks)

Luckily, Neo is flying at full speed to save her. Below is a slope field with Trinity’s altitude versus time.



Each unit in the vertical axis y corresponds to 1 floor and the units in the horizontal axis t are seconds.

- (a) (2 marks) In the movie, Neo manages to save Trinity. In the slope field above, sketch a possible trajectory for Trinity where she does not hit the ground. Make sure your sketch is clearly visible.
- (b) (2 marks) At what time does Neo arrive to save Trinity? Which features in the slope field allow you to identify this?

Time $\approx$

seconds

(c) (3 marks) This slope field is generated by a differential equation. Which of the following types of differential equations could give this slope field? (Select all options that apply)

☐ $\frac{dy}{dt} = f(t)$

☐ $\frac{dy}{dt} = g(y)$

☐ $\frac{dy}{dt} = h(t, y)$

Justify your answer below.

Q2. Consider the differential equation

$$\frac{dy}{dx} = y^2,$$

which of the following functions is a solution for this differential equation?

(1) $y(x) = \frac{1}{a-x}$

(2) $y(x) = e^x$

(3) $y(x) = \frac{x}{2-x}$

(4) $y(x) = x$

(5) $y(x) = \sin x$

Q3. Consider the slope field of the differential equation

$$\frac{dy}{dx} = y^3 - x^2$$

Along which of the following curves are the slopes constant?

(1) $y = x + 1$

(2) $y^2 = x^3 + 2$

(3) $y^3 = x^2 + 3$

(4) $3y^2 = 2x + 4$

Q6. Is it true that the integral

$$\int_1^{\infty} \frac{5 \sin(t^3 + 1)}{t^2} dt$$

converges?

Which of the following are improper integrals? Explain why.

☐ (a) $\int_1^{\infty} \frac{\sin x}{x} dx$

☐ (b) $\int_4^5 \frac{1}{x} dx$

☐ (c) $\int_3^4 \frac{1}{\sin x} dx$

☐ (d) $\int_{-3}^3 x^{-1/3} dx$

☐ (e) $\int_{-10}^{10} f(x) dx$ where $f(x) = \begin{cases} \frac{1}{x} + 2, & -10 \leq x < -1 \\ \frac{1}{x+2}, & -1 \leq x \leq 10 \end{cases}$

Calculate $\int_0^{\infty} x e^{-4x^2} dx$ if it converges. You may calculate the limit by appealing to the dominance of one function over another, or by l'Hopital's rule.

NOTE: Enter the exact answer or round to three decimal places.

If the integral does not converge, indicate that using the checkbox.

$\int_0^{\infty} x e^{-4x^2} dx =$ ☐ The integral does not converge.

3. Consider an arbitrary power series: $S(x) = c_0 + c_1(x - a) + c_2(x - a)^2 + c_3(x - a)^3 + \dots$

We say a power series **converges** at $x = b$, if $S(b)$ is equal to a finite number.

3.(a) $S(a)$ converges. What does it converge to?

3.(b) $S(x)$ can potentially converge at other values of x too. Either $S(x)$ converges on an interval centred at $x = a$ or it converges everywhere.

If $S(x)$ converges on an interval centred at a of length $2r$, what are the endpoints of that interval?

Hint: Draw a number line and count r units in each direction away from a .

3.(c) If the radius of convergence of $T(x) = \sum_{n=0}^{\infty} c_n(x + 3)^n$ is $r = 4$, what is the largest interval on which $T(x)$ converges?

3.(d) $T(x)$, from part (c), converges at which of the following: $x = -3$, $x = 3$, $x = 0$, $x = 8$, $x = 0.5$, $x = -0.5$, $x = -2$, $x = -8$, and/or $x = 10$?

7. Consider the predator-prey system:

$$\begin{cases} \frac{dy}{dt} = y - yx \\ \frac{dx}{dt} = -x + yx \end{cases}$$

Assume $x(0) = 1$ and $y(0) = 2$.

- (a) Determine which of the variables, x or y , represents the prey population and which represents the predator population. Explain.
- (b) Use the 1st degree Taylor polynomials centred at $t = 0$ to estimate $x(1)$ and $y(1)$.

$$x(t) \approx \quad , \quad x(1) \approx$$

$$y(t) \approx \quad , \quad y(1) \approx$$

- (c) Use the 1st degree Taylor polynomials centred at $t = 1$ to estimate $x(2)$ and $y(2)$.

$$x(t) \approx \quad , \quad x(2) \approx$$

$$y(t) \approx \quad , \quad y(2) \approx$$

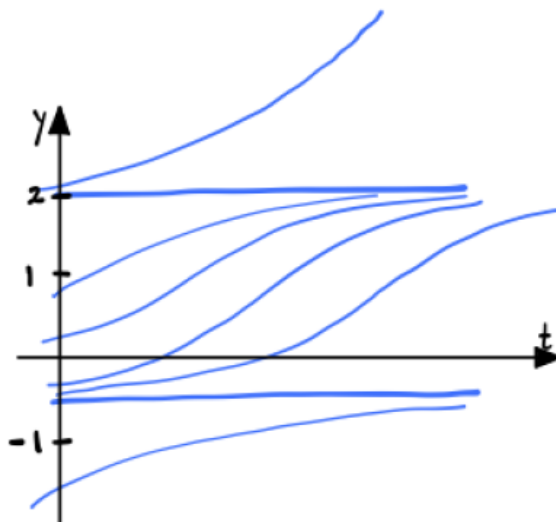
4 To which of the following scenarios could the equation

$$\frac{dT}{dt} = k(25 - T)$$

with solution $T = 25 - 20e^{-kt}$ correspond to?

- (A) A bottle of juice is bought from the store at 25 degrees and then it is placed in the fridge set at 5 degrees.
- (B) A bottle of juice is taken out of a 5 degree fridge and left in a 25 degrees room.
- (C) A bottle of juice is bought from the store at 20 degrees and then it is placed in the fridge set at 5 degrees.
- (D) A bottle of juice is taken out of a 20 degree fridge and left in a 25 degrees room.

11 Given the graph of solutions of a differential equation below, write a differential equation with such solutions.



$$\frac{dy}{dt} =$$

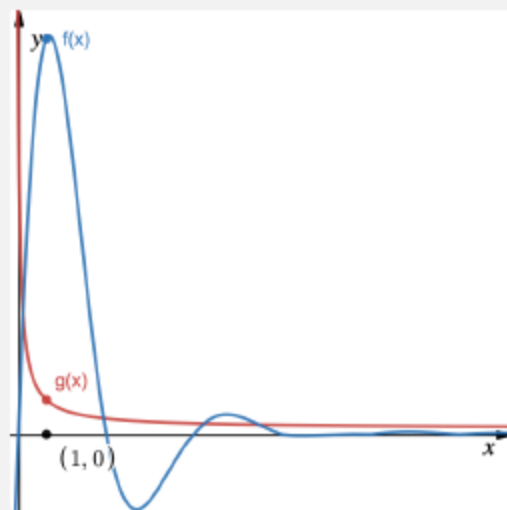
- 15 Newton's cooling equation for the temperature of a cup of coffee when its surroundings are at 20 degrees celsius is

$$\frac{dT}{dt} = k(20 - T)$$

Suppose that, initially, the cup is at 50 C and after one hour its temperature dropped by 10 C.

- (1) Find the equilibrium solutions of Newton's equation above and give a practical description of them.
- (2) Solve the differential equation with the conditions given in the statement above.
- (3) If we brew another cup of the same coffee to 100°C and leave it in the same room, is it possible for it to be at 80°C after one hour? Why / Why not?

Consider the functions $f(x)$ and $g(x)$.



- i) Suppose $\int_1^\infty f(x)dx < \infty$. What can you say about $\int_1^\infty g(x)dx$?
- ii) Suppose $\int_1^\infty f(x)dx = \infty$. What can you say about $\int_1^\infty g(x)dx$?
- iii) Suppose $\int_1^\infty g(x)dx < \infty$. What can you say about $\int_1^\infty f(x)dx$?
- iv) Suppose $\int_1^\infty g(x)dx = \infty$. What can you say about $\int_1^\infty f(x)dx$?

Four pairs of species are given, with descriptions of how they interact along with a number of systems of ODEs. Match the pairs of species with their corresponding system of ODEs.

1. Bees/flowers: each needs the other to survive.
2. Owls/ trees: owls need trees but trees are indifferent.
3. Elk/buffalo: in competition and would do fine alone;
4. Fox/hare: fox eats hare and needs it to survive.

$$(a) \begin{cases} \frac{dx}{dt} = -0.2x + 0.03xy \\ \frac{dy}{dt} = 0.4y - 0.08xy \end{cases}$$

$$(b) \begin{cases} \frac{dx}{dt} = 0.18x \\ \frac{dy}{dt} = -0.4y + 0.3xy \end{cases}$$

$$(c) \begin{cases} \frac{dx}{dt} = -0.6x + 0.18xy \\ \frac{dy}{dt} = -0.1y + 0.09xy \end{cases}$$

- (d) Write a possible system of ODEs for the species pair that does not have a match.

4. (11 marks) For each series, determine whether it is absolutely convergent, conditionally convergent or divergent. In each case, state the test you are using and justify your steps. This question continues on the next page.

(a)
$$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^3}$$

(b)
$$\sum_{n=1}^{\infty} \frac{n!}{n^n}$$

For the following series compute the radius R of convergence and the interval I of convergence. Justify your answer. Credit will only be given in so far as the reasons are adequate.

$$\sum_{n=2}^{\infty} (-1)^n \frac{x^n}{7^n \ln n}$$